ASSESSMENT TOOL 3

APPLICATION BASED QUESTIONS

Course Name	Cloud Computing and Big Data Analytics
Course Code	CSA15
Course Outcome	CO2 – Analyze virtualization architectures to identify performance and security issues in cloud computing environments (BL3)
Dave's Taxonomy	Manipulation (Level 2)
Question Count	5
Total Marks	25
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Programme / Branch	B.Tech – Artificial Intelligence and Machine Learning

CODE OF CONDUCT

I, **Paleru Dharani Govardhan**, certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: *Paleru Dharani Govardhan*

ASSESSMENT TASKS

Application Based Questions

1. Cloud Auto-Scaling and Load Balancing for Seasonal Sales

A company that experiences sudden increases in website traffic during seasonal sales can use cloud auto-scaling and load balancing to maintain performance. Auto-scaling automatically increases or decreases the number of computing resources according to demand. When the number of visitors increases, additional virtual machine instances or containers are started. When traffic becomes normal, unused instances are removed. This prevents the company from paying for resources that it does not need.

Load balancing distributes incoming user requests across multiple healthy servers. Instead of sending every request to one server, a load balancer shares the workload among several instances. If one server becomes overloaded or unhealthy, traffic can be redirected to other available servers. This reduces the risk of service interruption and improves response time.

Monitoring tools are important because they continuously observe CPU usage, memory, network traffic, request count, response time and error rates. Based on configured thresholds, monitoring services can trigger auto-scaling actions and alert administrators when unusual behaviour occurs.

For example, during a major online shopping sale, an e-commerce company may receive several times its normal traffic. Auto-scaling launches additional application instances, while a load balancer distributes customer requests among them. Monitoring detects rising traffic and helps maintain suitable capacity. As a result, customers can browse products and complete payments with fewer delays, while the company reduces downtime and provides a smoother shopping experience.

2. Secure Cloud Backup Storage, Durability and Disaster Recovery

Cloud storage services are designed to store large amounts of data with high durability and availability. Providers commonly maintain multiple copies of data across separate storage devices or facilities. This process is called data redundancy. If one disk or facility fails, another copy can be used to recover the information. Some services also provide versioning, which allows users to restore earlier versions of files after accidental deletion or modification.

Encryption protects backup data from unauthorized access. Data should be encrypted while being transferred over a network and while stored in the cloud. Encryption keys must be managed securely, and access should be restricted through identity and access management (IAM). Role-based access control gives each employee only the permissions required for their job. Multi-factor authentication and audit logs provide additional protection.

Cloud backup can be more cost-effective than maintaining traditional storage infrastructure. An organization does not need to purchase and maintain large storage servers, cooling systems or backup hardware in advance. It can pay according to the storage capacity and services used. Storage tiers also allow frequently accessed data and long-term archives to be stored at different costs.

For example, a hospital or business may back up its databases and important documents to cloud object storage. If a fire, hardware failure or ransomware incident affects its local data centre, the organization can retrieve clean backup copies from the cloud and restore its systems. This is a disaster recovery use case. Regular recovery testing is necessary to confirm that backups can actually be restored within the required time.

3. Content Delivery Networks and Global Application Performance

A Content Delivery Network (CDN) is a geographically distributed network of servers that delivers website content from locations closer to users. Static content such as images, videos, stylesheets and scripts can be

cached on CDN edge servers. When a user requests this content, it is delivered from a nearby edge location instead of always travelling to the application's main origin server. This reduces network distance, latency and loading time.

Edge servers improve user experience by serving cached content quickly and reducing the amount of traffic sent to the origin server. CDNs can also help handle sudden traffic spikes and provide protection against certain network attacks. Dynamic requests that cannot be cached are generally forwarded to the origin application or processed using suitable edge computing services.

Cloud providers support global availability through multiple regions, availability zones, global load balancing, managed databases and CDN services. Developers can deploy application components in more than one region and route users to suitable healthy endpoints. Replication and backup strategies help improve resilience, although they must be configured correctly.

For example, a global video-streaming application can store popular video segments at CDN edge locations. A viewer in India can receive content from a nearby edge server rather than downloading every segment from a distant server in another continent. Similarly, an online multiplayer game can use regional servers to reduce the distance between players and game infrastructure. These approaches reduce latency and make streaming and gameplay smoother.

4. Cloud Compliance, Auditing and Security in Financial Institutions

Financial institutions handle sensitive information such as customer identities, account details and transaction records. When they use cloud services, they must select providers and configurations that support the regulatory requirements applicable to their operations. Cloud providers may offer compliance programmes, security certifications, contractual controls and documentation that help customers assess the service. However, using a compliant provider does not automatically make the customer's application compliant. The institution remains responsible for configuring and operating its workloads correctly.

Logging records activities such as user sign-ins, administrative changes, database access and API requests. Monitoring analyses these events and can identify unusual patterns, such as repeated failed logins or access from unexpected locations. Audit trails help investigators understand who accessed a resource and what action was performed. Logs should be protected from unauthorized modification and retained according to the institution's policies and legal requirements.

Encryption protects data both at rest and in transit. Strong identity management, least-privilege access, multi-factor authentication, network segmentation and secure key management further reduce security risks. Regular vulnerability assessments and security audits help identify weaknesses.

Data sovereignty means that data is stored and processed in accordance with the laws and requirements of the relevant jurisdiction. The institution should select appropriate cloud regions, review data-location commitments,

configure backups carefully and verify where replicas and support operations may process data. Legal and compliance teams should review the provider's terms.

For example, a bank processing online fund transfers can encrypt transaction data, restrict database access to authorized services, and record every privileged action. Security monitoring can flag suspicious access, while audit logs support investigations. The bank should also ensure that customer data and backups remain in approved locations where required by applicable regulations.

5. Containers, Virtual Machines and Kubernetes Orchestration

A virtual machine (VM) runs a complete guest operating system on virtualized hardware managed by a hypervisor. Each VM has its own operating system kernel and allocated virtual CPU, memory and storage. This provides strong isolation, but VMs usually require more resources and take longer to start than containers.

A container packages an application with its libraries and dependencies while sharing the host operating system kernel. Containers are generally lightweight, start quickly and allow more application instances to run on the same hardware. They are useful for deploying applications consistently, although container security depends on correct image management, isolation, permissions and host configuration.

Container orchestration platforms such as Kubernetes automate the management of containerized applications. Kubernetes can schedule containers across cluster nodes, restart failed containers, perform health checks, expose services through networking, and scale application replicas according to demand. Rolling updates help teams release new versions with reduced disruption. These capabilities simplify deployment and day-to-day operations.

Containers also improve portability. A container image built using a standard format can run in different environments that support the container runtime, including local development systems, private data centres and public cloud platforms. This reduces environment-related differences and helps avoid dependence on one specific infrastructure provider. However, networking, storage, identity and managed-service integrations may still need environment-specific configuration.

For example, an online food-delivery application may use separate microservices for user accounts, restaurant listings, orders, payments and delivery tracking. Each service can run in its own container and be updated independently. Kubernetes can scale the order service during busy meal times while keeping other services at their normal capacity. This improves resource utilization, deployment flexibility and application scalability.

ASSESSMENT RUBRIC

Criteria	Weightage	Level 4 – Exemplary
Relevance to Application	25%	Directly addresses the given use

		case with strong relevance.
Use of Cloud Concepts	25%	Applies appropriate cloud concepts accurately.
Practical Insight	20%	Shows strong real-world understanding.
Completeness	15%	Covers all required aspects of the question.
Clarity	15%	Presents information in a clear and concise manner.